آغا خان یونیورسٹی ایگزامینیشن بورڈ

AGA KHAN UNIVERSITY EXAMINATION BOARD

Notes from E-Marking Centre on HSSC-II Physics Annual Examinations 2024

Introduction

This document has been produced for the teachers and candidates of Higher Secondary School Certificate (HSSC) Part II Physics. It contains comments on candidates' responses to the 2024 HSSC-II Examination indicating the quality of the responses and highlighting their relative strengths and weaknesses.

E-Marking Notes

This includes overall comments on candidates' performance on every question and *some* specific examples of candidates' responses that support the mentioned comments. Please note that the descriptive comments represent an overall perception of the better and weaker responses as gathered from the e-marking session. However, the candidates' responses shared in this document represent some specific example(s) of the mentioned comments.

Teachers and candidates should be aware that examiners may ask questions that address the Student Learning Outcomes (SLOs) in a manner that requires candidates to respond by integrating knowledge, understanding and application skills they have developed during the course of study. Candidates are advised to read and comprehend each question carefully before writing the response to fulfil the demand of the question.

Candidates need to be aware that the marks allocated to the questions are related to the answer space provided on the examination paper as a guide to the length of the required response. A longer response will not in itself lead to higher marks. Candidates need to be familiar with the command words in the SLOs which contain terms commonly used in examination questions. However, candidates should also be aware that not all questions will start with or contain one of the command words. Words such as 'how', 'why' or 'what' may also be used.

General Observations

Generally, candidates demonstrated a strong understanding of concepts including carbon-coded resistors, electromagnetism and electromagnetic induction, the reactance of an inductor, magnetic properties of solids, electronics, X-rays and nuclear radiation. However, they struggled with questions based on modern physics and numerical problems. Therefore, teachers must focus on these areas and offer additional drills and practice to help candidates develop a thorough understanding.

- Recognise the importance of unit conversion and extracting data from the question.
- Identify the demand of the question and application of the correct formula with correct substitution and calculations.
- Understand the keywords and command words used in the question.

Note: Candidates' responses shown in this report have not been corrected for grammar, spelling, format, or information.

DETAILED COMMENTS

Constructed Response Questions (CRQs)

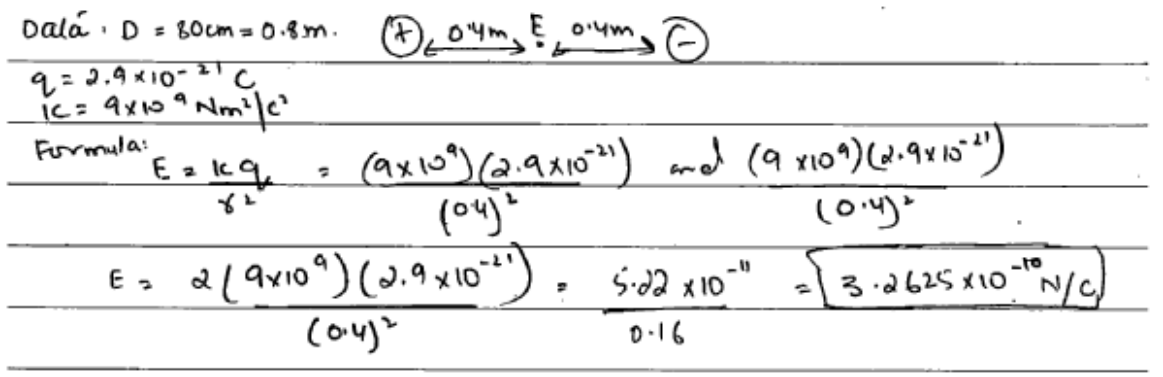
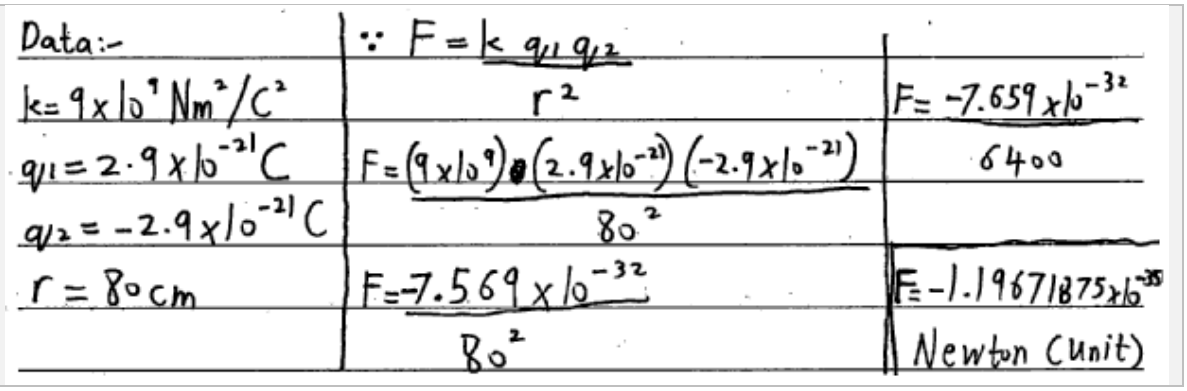
Question No. 1	
Question Text	Two-point charges 2.9×10^{-21} C and -2.9×10^{-21} C are separated by a distance of 80 cm. Calculate the magnitude of electric field intensity at the midpoint of the charges. (**Note**: Take the value of the constant of proportionality 'k' as 9×10^9 Nm^2/C^2.)
SLO No.	11.3.2
SLO Text	Derive an expression for the magnitude of electric field of a distance or from a point charge "q" and use the expression in solving word problems.
Max Marks	03
Cognitive Level	A*
Checking Hints	1 mark for converting 'cm' to 'm' 1 mark for writing the correct formula/ substituting the values 1 mark for writing the correct answer with the SI unit
Overall Performance	The candidates' overall performance on this question was generally satisfactory, demonstrating strong computational skills and an understanding of electric fields by point charges. Their ability to identify key factors in the question reflects their solid analytical skills. However, some candidates missed applying the superposition principle when calculating the net electric field, indicating a gap in their understanding of all aspects of the question.
Description of Better Responses	*Better responses* reflected candidates' proficiency in extracting relevant data from the question and using the correct formula i.e., $E = k\,q / r^2$ to calculate electric field intensity at the midpoint between the two charges. These responses demonstrated a solid understanding of the concept of the electric field and the correct application of the principle of superposition. Many candidates provided step-by-step calculations, leading to correct results as shown in the image of better response. These responses indicated a strong grasp of the physical concepts and the mathematical procedures involved.
Image of Better Response	
Description of Weaker Responses	In *weaker responses*, candidates misunderstood the question and calculated the force between the charges instead of determining the electric field intensity at the midpoint. This confusion between the concepts of electric field intensity and electrostatic force was evident. Additionally, some candidates calculated the electric field intensity due to only one charge, overlooking the contribution from the second charge. This mistake often results in incorrect answers and underscores the importance of considering all relevant factors in such problems.

Image of Weaker Response	

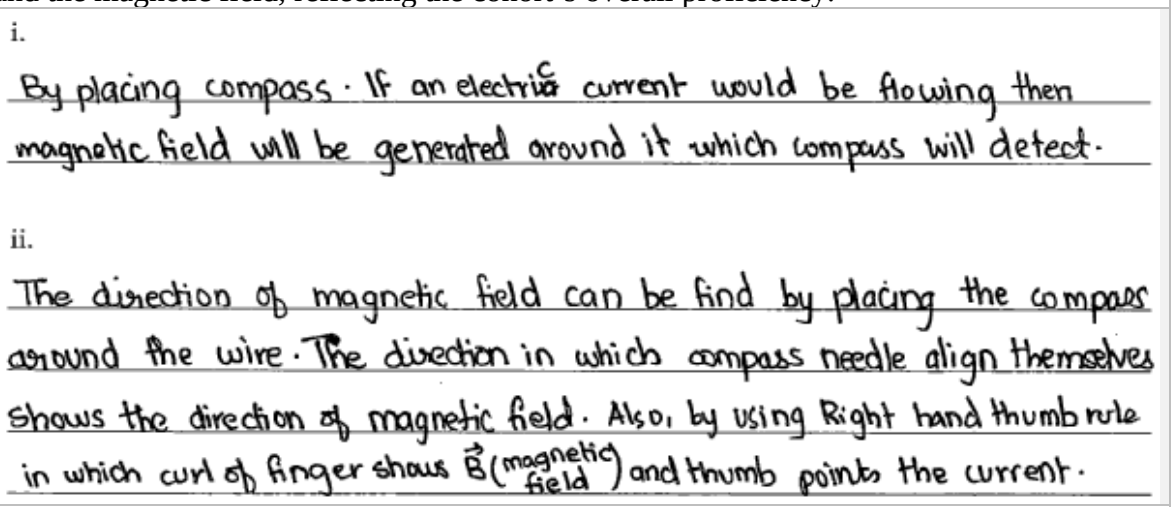
Suggestions for improvement (Highlight all that apply)

Maximising SLO Achievement	**Preferred Pedagogy** Used for this SLO**	**Assessment Strategies**
• Identify the expectation of command words (use Command Word Guide) • Ensure the content is taught at the relevant cognitive level • Identify necessary content required (skills + concepts) • Review past paper questions on the concept • Utilise the resource guide for additional materials	• Story Board • Cause and Effect • Fish and Bone • Concept Mapping • Audio Visual Resources • Think, Pair and Share • Knowledge Platform videos • Questioning Technique (Socratic approach) • Practical Demonstration ** For description of each Pedagogy, refer to Annexure A	• Past paper questions • Discussion on E-Marking Notes • AKU-EB Digital Learning Solution powered by Knowledge Platform https://akueb.knowledgeplatform.com/login

Any Additional Suggestion:

- Teachers are encouraged to discuss the concepts of electrostatic force and electric field intensity, highlighting the differences between electric force and electric field. Emphasising the conditions under which each concept applies and the correct formulas to use will help prevent similar mistakes in the future.
- Encouraging continued practice and application of concepts in different contexts will further enhance candidates' mathematical approach toward problem-solving abilities and ensure continued success in similar scenarios.
- Subject teachers must encourage candidates to carefully read and understand the full scope of the question is crucial. Misinterpretations can be reduced by reinforcing the need to analyse all aspects of the question before beginning calculations.

*K = Knowledge U = Understanding A = Application and other higher-order cognitive skills

Question No. 2	
Question Text	A copper wire is hidden beneath a carpet. How can we securely determine the following? i. Any electrical current flowing through the copper wire ii. The direction of magnetic field produced by the electrical current
SLO No.	13.1.2
SLO Text	Describe the direction of magnetic field produced by a current carrying conductor.
Max Marks	03
Cognitive Level	U
Checking Hints	1 mark for writing the highlighted statement in part (i) 1 mark for writing each highlighted statement in part (ii) (2 required)
Overall Performance	Overall, the cohort demonstrated a thorough understanding of the principles involved in detecting current flow in a wire and determining the direction of the magnetic field. Responses indicated a good grasp of the relationship between electric current and the magnetic field. However, placing further emphasis on selecting accurate and complete methods will improve the clarity and accuracy of responses.
Description of Better Responses	In *better responses*, many candidates provided well-reasoned answers for both parts of the question. In the first part, candidates accurately identified that a compass can be used to detect the presence of a current, explaining that the deflection of the compass needle indicates a magnetic field and thus confirms current flow through the wire. In the second part, candidates effectively described the use of the right-hand rule or the observation of compass needle deflection to determine the direction of the magnetic field. These insightful explanations demonstrate a strong understanding of the relationship between electric current and the magnetic field, reflecting the cohort's overall proficiency.
Images of Better Responses	
Description of Weaker Responses	In *weaker responses*, some candidates provided incorrect or vague justifications, such as suggesting the use of methods not relevant to the problem or failing to explain the connection between current flow and magnetic field detection. They wrote about different conductors like copper, iron, silver and their conduction properties. A few responses lacked clarity in explaining the proper use of the compass or the right-hand rule, indicating a partial understanding of the principle involved.

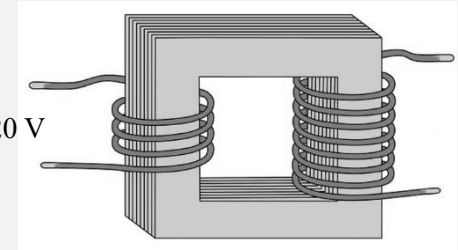
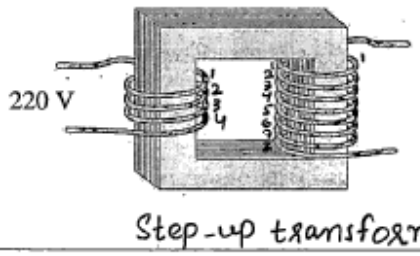
Images of Weaker Responses	i. Copper is a good conductor and due to a conductor the electrical current flow through it. ii. The flowing of electrical current also produced a magnetic field and determined the direction of magnetic field and current.

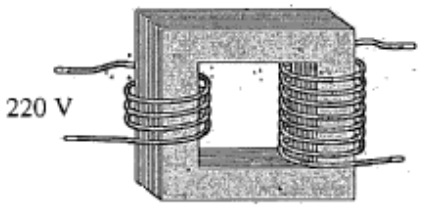
Suggestions for improvement (Highlight all that apply)

Maximising SLO Achievement	Preferred Pedagogy Used for this SLO	Assessment Strategies
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Any Additional Suggestion:

- Teachers are encouraged to build a solid foundation for their students in understanding the magnetic field produced by current-carrying conductors and the Biot-Savart law. It is important to include practical demonstrations and hands-on experiences in these topics as part of classroom instruction.
- Teachers are encouraged to use online tools and simulations that allow students to interactively experiment with electric current and analyse its effects.
- Regular practice and application of magnetic field intensity concepts in various scenarios can further reinforce candidates' comprehension and improve their performance in similar questions.
- Reinforcing the concepts of magnetic field detection and direction determination using a compass and the right-hand grip rule can help candidates provide more precise and accurate responses.
- Encouraging candidates to clearly articulate their reasoning and ensure that their methods are scientifically sound will improve the quality of their explanations.

Question No. 3	
Question Text	i. Write the name of the given transformer.  ii. Calculate the voltage in the secondary coil.
SLO No.	14.5.4
SLO Text	Derive Ns/Np = Vs/Vp and Vs.Is = Vp.Ip for an ideal transformer and use it for solving word problems.
Max Marks	03
Cognitive Level	A
Checking Hints	1 mark for naming the transformer 1 mark for writing the correct formula/ substitution 1 mark for writing the correct answer with the SI unit
Overall Performance	The overall performance of the cohort on this question was highly commendable. The majority of the candidates demonstrated a good understanding of the construction of a transformer. They accurately identified the transformer as a 'step-up transformer' and calculated the output voltage by using a transformation equation (Ns/Np = Vs/Vp). Overall, the answers reflected a strong grasp of the transformer's construction, indicating a sound knowledge of the topic.
Description of Better Responses	In *better responses*, candidates correctly recognised the key components of a given transformer and identified it as a step-up transformer. They accurately counted the number of turns in the primary and secondary coils, demonstrating a good understanding of transformer basics. For calculating the secondary voltage, a transformation equation (Ns/Np = Vs/Vp) of a transformer was used that provided the correct and precise answer. Candidates' understanding of the transformer's construction and its working was evident, reflecting their grasp of its operation.
Image of Better Response	i.  Step-up transformer. ii. $\frac{Vp}{Vs} = \frac{Np}{Ns}$ → 220 × 2 = Vs Vs = 440V $\frac{220}{Vs} = \frac{4 1}{8 2}$ Data: Vp = 220V Np = 4 coils Ns = 8 coils

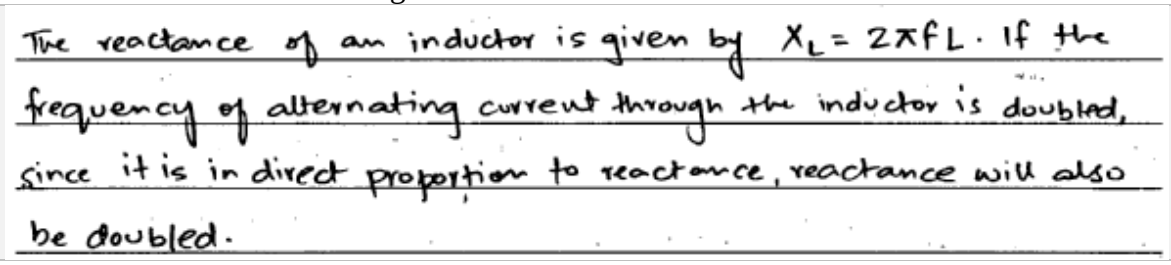
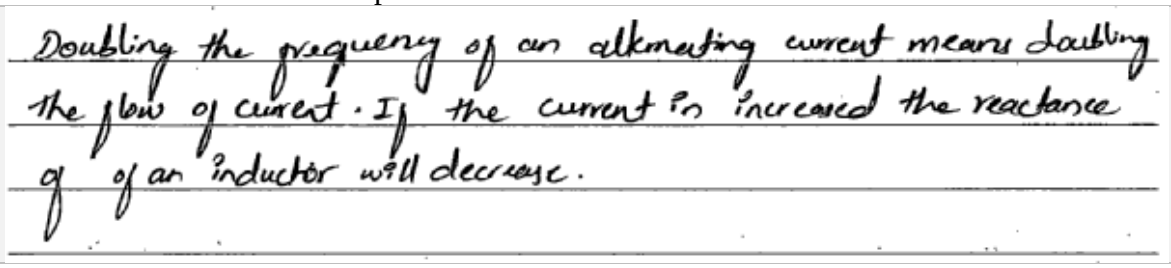
Description of Weaker Responses	In *weaker responses*, candidates provided improper or incorrect identification of the type of transformer, indicating a lack of clarity regarding the transformer's construction, possibly confusing the step-up transformer with a step-down transformer, despite a clear indication from the coil turns and the input and output voltage relationship. Some responses contained errors in calculating secondary voltage due to incorrect application of the transformation equation, they used Vs.Is = Vp.Ip or misunderstandings about the relationship between coil turn and voltage ratio.
Image of Weaker Response	i.  step down transformer. ii. 220 ÷ 3 = 73. 73 x 7 = 511V. = 511 V Ans (secondary coil).

Suggestions for improvement (Highlight all that apply)

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Any Additional Suggestion:

- Teachers are encouraged to incorporate visual diagrams and animations of electrical instruments during their classroom teaching.
- Encouraging candidates to reinforce their understanding of the fundamental principles governing transformers, including the relationship between coil turns and voltage levels, is essential. A solid grasp of these basic formulas and concepts will help reduce calculation errors in future assessments.

Question No. 4	
Question Text	How does doubling the frequency of an alternating current affect the reactance of an inductor?
SLO No.	15.2.1
SLO Text	Explain the flow of AC through resister, capacitor and inductor.
Max Marks	02
Cognitive Level	U
Checking Hints	1 mark for writing the correct formula of an inductor 1 mark for writing the correct answer of an inductor
Overall Performance	The overall performance on this question was strong, reflecting a solid understanding of the relationship between frequency and inductive reactance. Many explained their answers by referencing the formula for inductive reactance i.e., $X_L = 2 \pi f L$. However, some candidates confused inductive reactance with capacitive reactance, leading to incorrect answers. There is a need for clearer explanations and better differentiation between these similar concepts.
Description of Better Responses	In *better responses*, candidates effectively recognised that the reactance of an inductor increases with the frequency of the alternating current. They accurately used the formula $X_L = 2 \pi f L$. Additionally, better responses not only offered accurate mathematical explanations but also explored the physical implications of increased reactance, such as a decrease in current flow through the inductor.
Image of Better Response	
Description of Weaker Responses	In *weaker responses*, candidates demonstrated confusion between inductive and capacitive reactance, leading to incorrect explanations. Some responses lacked clarity in explaining why reactance changes with frequency, indicating a superficial understanding of the underlying principles. They wrote that doubling or tripling the frequency the alternating current will also double or triple and if the current increases the reactance will also increase.
Image of Weaker Response	

Suggestions for improvement (Highlight all that apply)

Maximising SLO Achievement	Pedagogy Used for that SLO	Assessment Strategies
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Any Additional Suggestion:

- Reinforcing the distinction between inductive and capacitive reactance and their respective dependencies on frequency will help candidates provide accurate explanations.
- Emphasising the importance of using correct terminology and clearly articulating the relationship between frequency and reactance will improve the candidates' response quality.

Question No. 5	
Question Text	Define the following terms. i. Paramagnetism ii. Ferromagnetism
SLO No.	16.5.2
SLO Text	Describe diamagnetic, paramagnetic and ferromagnetic solids.
Max Marks	02
Cognitive Level	K
Checking Hints	1 mark for each correct definition (2 required)
Overall Performance	The overall performance on this question was average. Many candidates struggled to provide clear and accurate definitions, with some confusing the properties of paramagnetic substances with those of ferromagnetic substances. While some candidates demonstrated a solid understanding of the concept and offered clear, accurate definitions, others gave vague or incomplete answers.
Description of Better Responses	In *better responses*, candidates provided clear definitions of the paramagnetic i.e., such substances in which the orbits and spin axis of the electrons in an atom are so adjusted that their fields support each other and the atoms behave like a tiny magnet. They also wrote correct definition of ferromagnetic i.e., such substances in which the atoms cooperate in such a way to exhibit strong magnetic field.

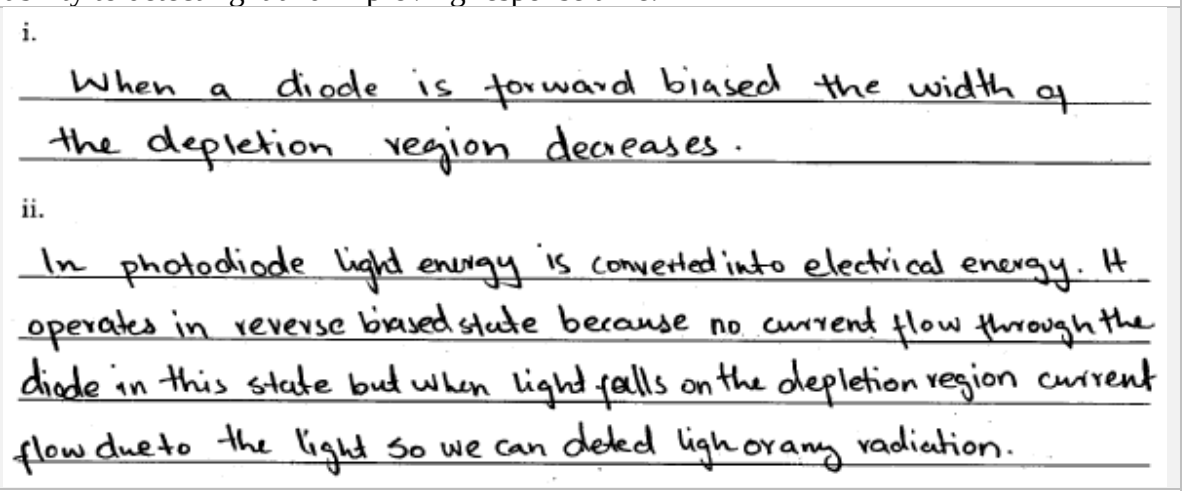
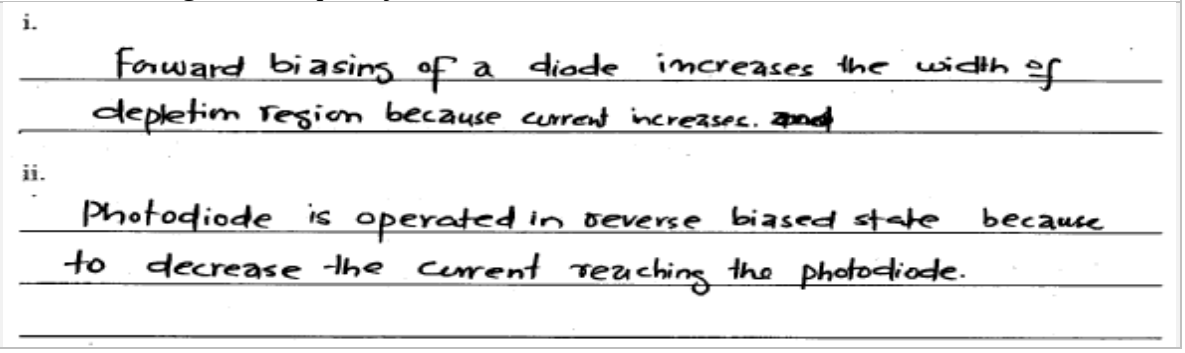
Image of Better Response	i. Magnetism which occurs when the spin and orbital motion of electrons are add up to produce magnetism in a substance. ii. A phenomenon in which a substance is put under the influence of external magnetic field and the max no. of electron in domain get aligned in the same direction producing a strong magnetism
Description of Weaker Responses	In *weaker responses,* candidates either provided incomplete or inaccurate definitions or mentioned irrelevant concepts and wrote about poles of magnetic materials and some wrote about positive and negative charges and struggled to differentiate the characteristics of paramagnetic and ferromagnetic materials.
Image of Weaker Response	i. paramagnetism is aprocess in which a magnet acts as only one pole, N or S. Haf magnet. ii. in ferromagnetism a magnet is divided into two poles "N" and "S" having +ve and -ve charge

Suggestions for improvement (Highlight all that apply)

Maximising SLO Achievement	**Pedagogy Used for that SLO**	**Assessment Strategies**
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Any Additional Suggestion:

- Emphasising the distinct characteristics of paramagnetic and ferromagnetic substances in teaching materials and classroom discussions can help improve clarity.
- Candidates should be encouraged to focus on the definitions and properties of the magnetic materials included in the topic.
- Providing more examples and practical applications by the teachers on these materials could aid in reinforcing their understanding and ability to distinguish between different types of magnetic substances.

Question No. 6	
Question Text	i. What is the effect of forward biasing of a diode on the width of depletion region? ii. Why is a photodiode operated in reverse biased state? Give TWO reasons.
SLO No.	17.2.9
SLO Text	Describe the function and uses of light-emitting diodes (LEDs) and photodiodes.
Max Marks	03
Cognitive Level	U
Checking Hints	1 mark for writing the depletion region as 'narrowed' 1 mark for writing each reason (Any 2 required)
Overall Performance	The overall performance on this question was good. Most candidates accurately described the effect of forward biasing on the depletion region. However, the second part of the question was less well-understood, resulting in a variety of responses and requiring further refinement.
Description of Better Responses	In the first part of the question, candidates correctly described the effect of forward biasing on the depletion region of a diode. They accurately explained that the forward biasing reduces the width of the depletion region, allowing current to flow more easily through the diode. Many responses also included a proper justification, detailing how the applied voltage decreases the potential barrier and facilitates charge carrier movement across the junction. In the second part of the question, candidates correctly noted that reverse biasing increases the depletion region, improving the photodiode's sensitivity to light enhancing the ability to detect light and improving response time.
Image of Better Response	 i. When a diode is forward biased the width of the depletion region decreases. ii. In photodiode light energy is converted into electrical energy. It operates in reverse biased state because no current flow through the diode in this state but when light falls on the depletion region current flow due to the light so we can deted ligh or any radiation.
Description of Weaker Responses	In *weaker responses*, candidates often provided statements without thorough explanations or justifications. Many candidates confused forward and reverse biasing, mixing up the concepts and making unsupported statements. In the second part of the question, many candidates incorrectly suggested that reverse biasing a photodiode was intended to prevent it from conducting or confused the operation of a photodiode with that of other types of diodes, leading to inadequate justifications.
Image of Weaker Response	 i. Forward biasing of a diode increases the width of depletim region because current increases. and ii. Photodiode is operated in reverse biased state because to decrease the current reaching the photodiode.

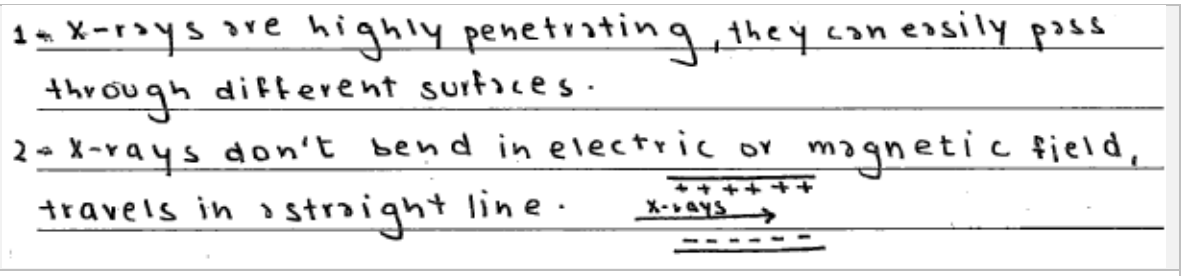
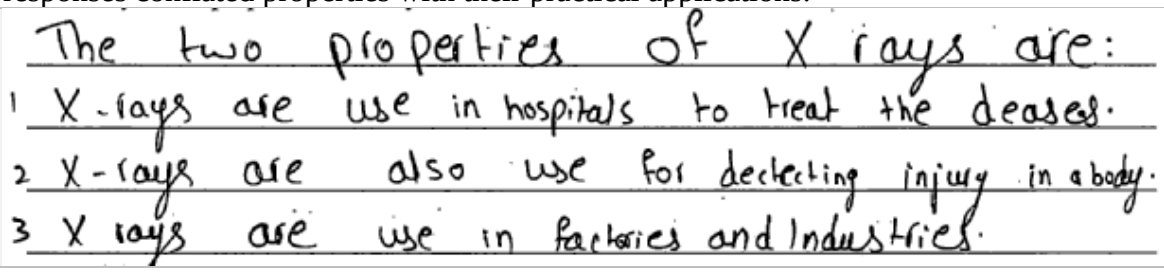
Suggestions for improvement (Highlight all that apply)

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Any Additional Suggestion:

- Focusing on the physical and functional reasons for these configurations will aid in improving the quality of responses.
- Strengthening the understanding of the specific roles of forward and reverse biasing in different types of diodes is necessary to score higher marks in such questions.

Question No. 7	
Question Text	Write any TWO properties of X-rays.
SLO No.	19.4.3
SLO Text	Explain the production, properties and uses of X-rays.
Max Marks	02
Cognitive Level	K
Checking Hints	1 mark for writing each property (Any 2 required)
Overall Performance	The overall performance of the cohort on this question was highly commendable but there is room for improvement in clearly distinguishing between properties and uses. The majority of candidates demonstrated a clear understanding of the question's requirements and effectively showcased their knowledge of the fundamental characteristics of X-rays. However, some candidates confused the properties of X-rays with their applications. There was also confusion between X-rays and other types of electromagnetic radiation, leading to incorrect or incomplete property listings.
Description of Better Responses	In *better responses*, candidates effectively addressed the question's requirements, demonstrating a clear understanding of X-ray properties. They provided accurate and well-defined characteristics of X-rays, including their high frequency and short wavelength, their ability to penetrate materials, their ionising nature, their lack of deflection by electric and magnetic fields and their movement at the speed of light.

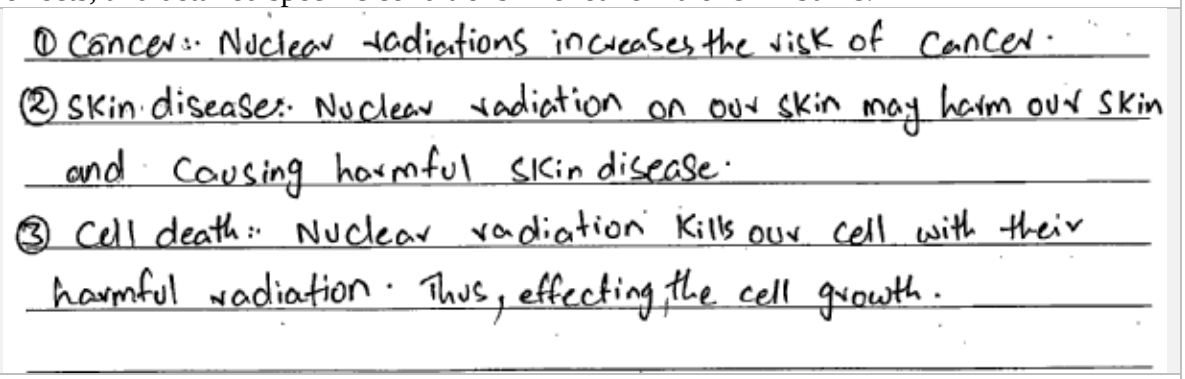
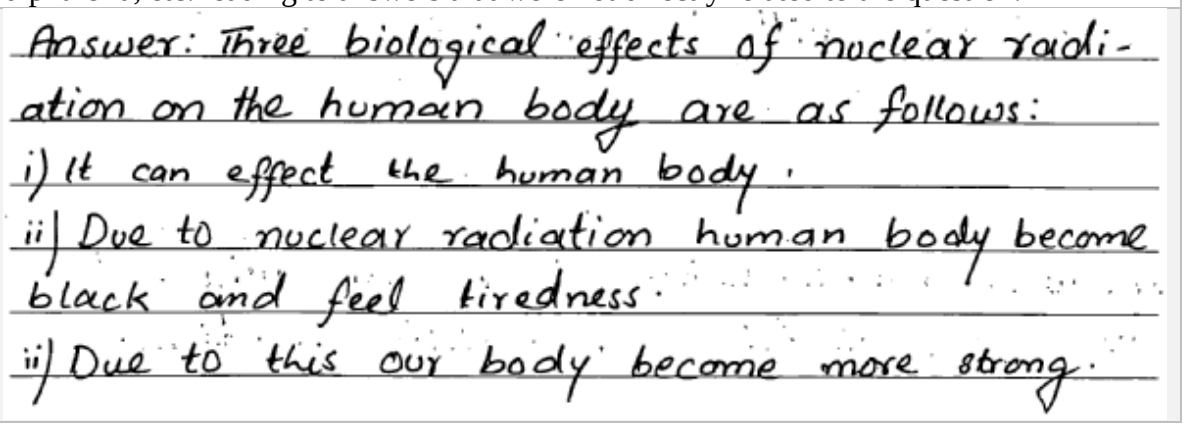
Image of Better Response	
Description of Weaker Responses	In *weaker responses,* candidates struggled to effectively differentiate between the properties and uses of X-rays. Many responses listed applications or uses of X-rays, such as medical imaging or dental X-rays, as properties. Instead of describing the intrinsic physical characteristics of X-rays, such as their high frequency and penetrating ability, these responses conflated properties with their practical applications.
Image of Weaker Response	

Suggestions for improvement (Highlight all that apply)

Maximising SLO Achievement	**Pedagogy Used for that SLO**	**Assessment Strategies**
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Any Additional Suggestion:

- Encouraging candidates to focus on the fundamental physical aspects of X-rays, such as wavelength, frequency and interactions with matter will enhance their ability to accurately describe these properties.
- Teachers are suggested to clarify the distinction between the inherent properties of X-rays (their intrinsic characteristics) and their applications (how they are utilised in various fields) will help their students prevent this confusion.

Question No. 8	
Question Text	Describe any THREE biological effects of nuclear radiations on the human body.
SLO No.	20.10.1
SLO Text	Discuss the biological effects due to exposure of nuclear radiations.
Max Marks	03
Cognitive Level	U
Checking Hints	1 mark for writing each effect (Any 3 required)
Overall Performance	Overall, candidates addressed this question effectively, displaying a strong understanding of the effects of nuclear radiation on the human body. Their well-informed responses reflect a solid grasp of the biological impacts of nuclear radiation.
Description of Better Responses	In *better responses*, the majority of the candidates provided well-informed responses, correctly identifying various biological effects of nuclear radiation on the human body. Commonly mentioned effects included radiation sickness, an increased risk of cancer and damage to tissues and organs. Many candidates demonstrated a clear understanding of the severity and types of effects depending on the radiation dose, such as acute and chronic effects, and detailed specific conditions like leukemia or skin burns.
Image of Better Response	
Description of Weaker Responses	In *weaker responses*, candidates encountered challenges in meeting the question's demand. Some responses lacked specificity or detail, merely stating harmful effects without elaborating on the specific biological impacts. A few candidates confused the biological effects with the physical properties of radiation, such as skin colour, tiredness, nausea and diphtheria, etc. leading to answers that were not directly related to the question.
Image of Weaker Response	

Suggestions for improvement (Highlight all that apply)

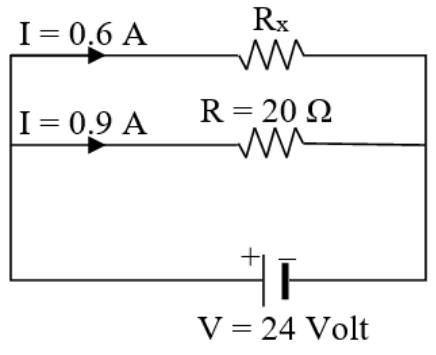
Maximising SLO Achievement	Pedagogy Used for that SLO	Assessment Strategies
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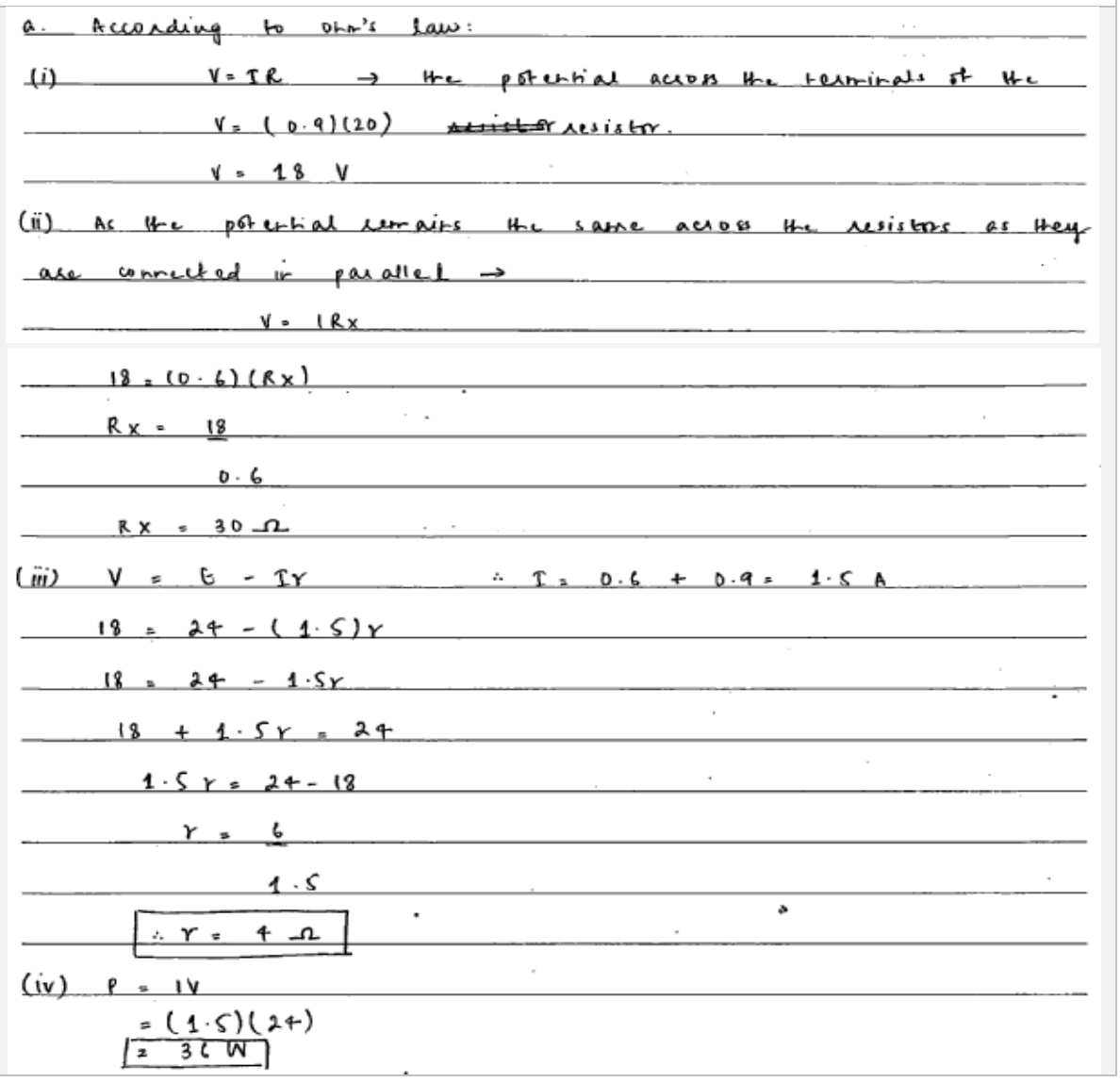
Any Additional Suggestion:

- Encouraging students to provide specific examples and explanations of biological effects can enhance the depth and quality of their responses.
- Teachers are advised to make understanding the mechanisms through which radiation affects the body, such as DNA damage and cellular mutations. Reinforcing the importance of specificity in response, distinguishing between general harm and specific biological effects will improve their students' clarity and accuracy.
- Further emphasis on providing detailed and specific examples will continue to strengthen student performance in this area.

Extended Response Questions (ERQs)

Extended response questions offered a choice between parts 'a' and 'b'

Question No. 9a	
Question Text	The given circuit diagram shows two resistors are connected in parallel with a battery.  Calculate the following. i. The terminal velocity of the circuit. ii. The unknown resistance (Rx). iii. The internal resistance of the battery. iv. The power delivered by the battery.

SLO No.	12.4.6
SLO Text	Calculate the power dissipation due to the internal resistance of a circuit.
Max Marks	07
Cognitive Level	A
Checking Hints	1 mark for writing the correct formula of Ohm's law 1 mark for the correct value 'V' 1 mark for the correct value 'R' 1 mark for the correct formula of the internal resistance 1 mark for the correct value of internal resistance 1 mark for the correct formula of the power dissipation 1 mark for the correct value of power dissipation with SI unit
Overall Performance	This part of the question was least attempted by candidates. However, most responses showed good circuit analysis skills and a solid understanding of the concepts. However, there is a need for clearer differentiation between e.m.f. and terminal voltage.
Description of Better Responses	In *better responses*, candidates demonstrated strong circuit analysis skills by correctly calculating terminal voltage by using Ohm's law (V = I R). They effectively used this terminal voltage (V = E – I r) to determine the value of unknown resistance by using r = [(E - V) / I] and applied Kirchhoff's first law to find the total current in the circuit. Many also accurately calculated the internal resistance of the battery and the power (P = I V) delivered by it, showing a solid understanding of the relationships between voltage, current, and resistance.
Image of Better Response	

a. According to Ohm's law:

(i) $V = IR$ → the potential across the terminals of the resistor.

$V = (0.9)(20)$

$V = 18\ V$

(ii) As the potential remains the same across the resistors as they are connected in parallel →

$V = IRx$

$18 = (0.6)(Rx)$

$Rx = \frac{18}{0.6}$

$Rx = 30\ \Omega$

(iii) $V = E - Ir$ $\quad \therefore I = 0.6 + 0.9 = 1.5\ A$

$18 = 24 - (1.5)r$

$18 = 24 - 1.5r$

$18 + 1.5r = 24$

$1.5r = 24 - 18$

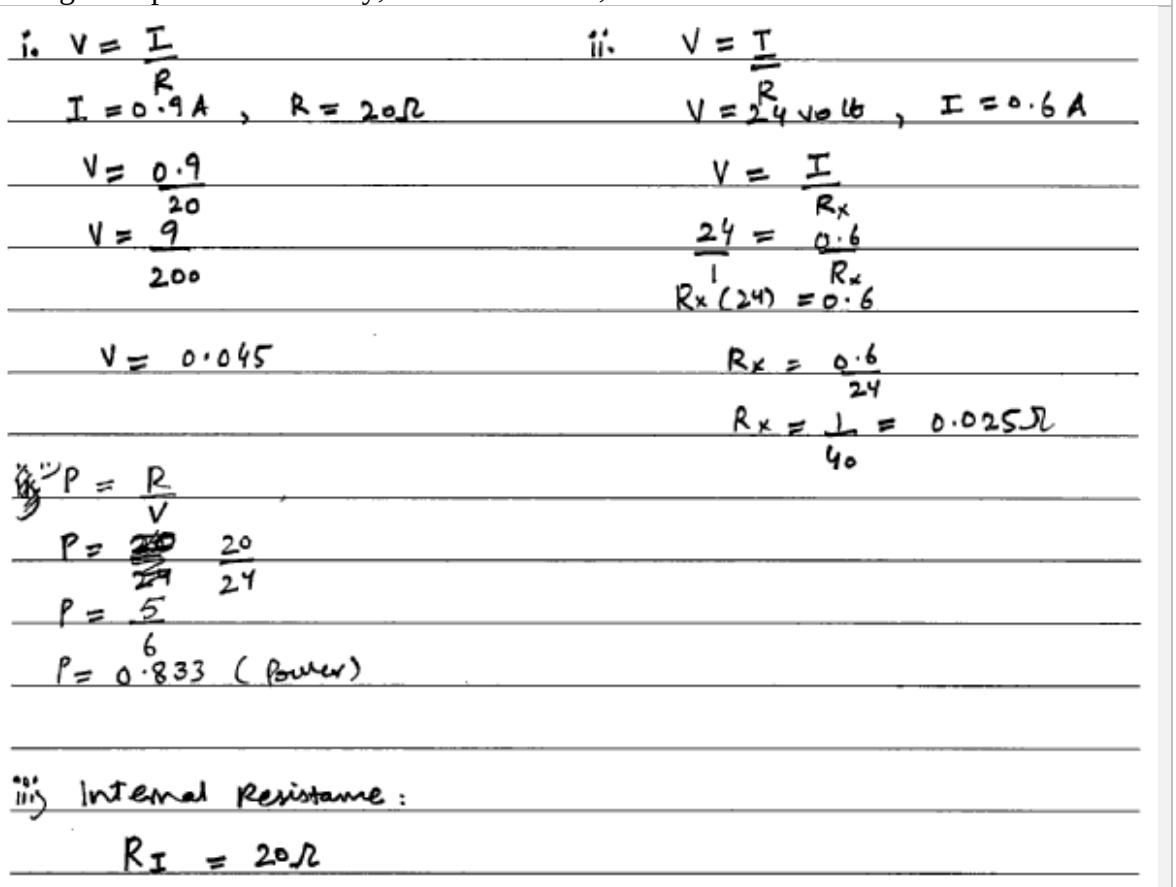
$r = \frac{6}{1.5}$

$\therefore r = 4\ \Omega$

(iv) $P = IV$

$= (1.5)(24)$

$= 36\ W$

Description of Weaker Responses	In *weaker responses*, candidates incorrectly used the electromotive force (e.m.f.) instead of the terminal voltage to calculate the unknown resistor, leading to incorrect results. This error halted from the confusion between the e.m.f. and terminal voltage, the latter accounting for the battery's internal resistance. Some students showed difficulties in applying Kirchhoff's laws, leading to errors in calculating the total current and subsequently the internal resistance and power. Most of the responses wrote all the formulas of Ohm's law, terminal voltage and power incorrectly, like as V = I / R, P = R / V etc.
Image of Weaker Response	

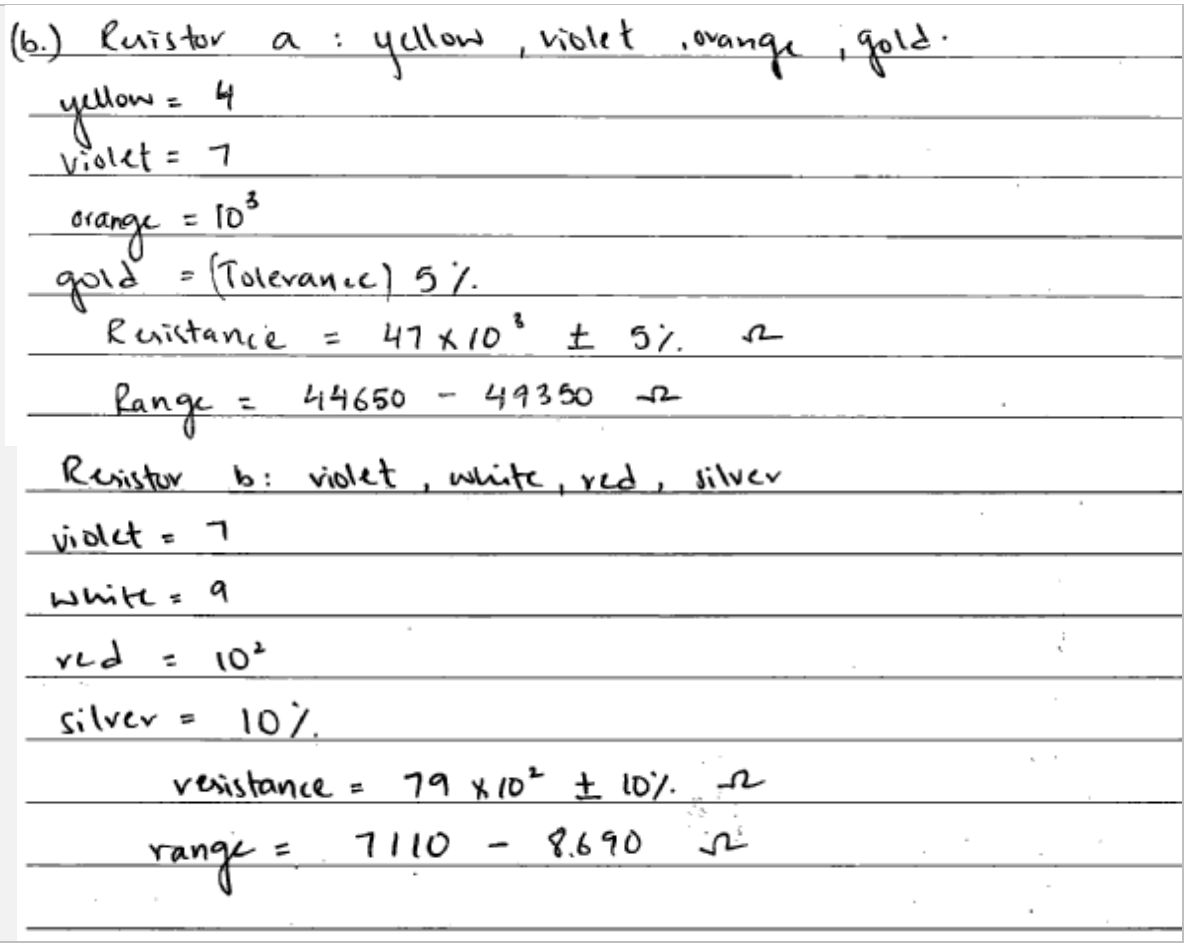
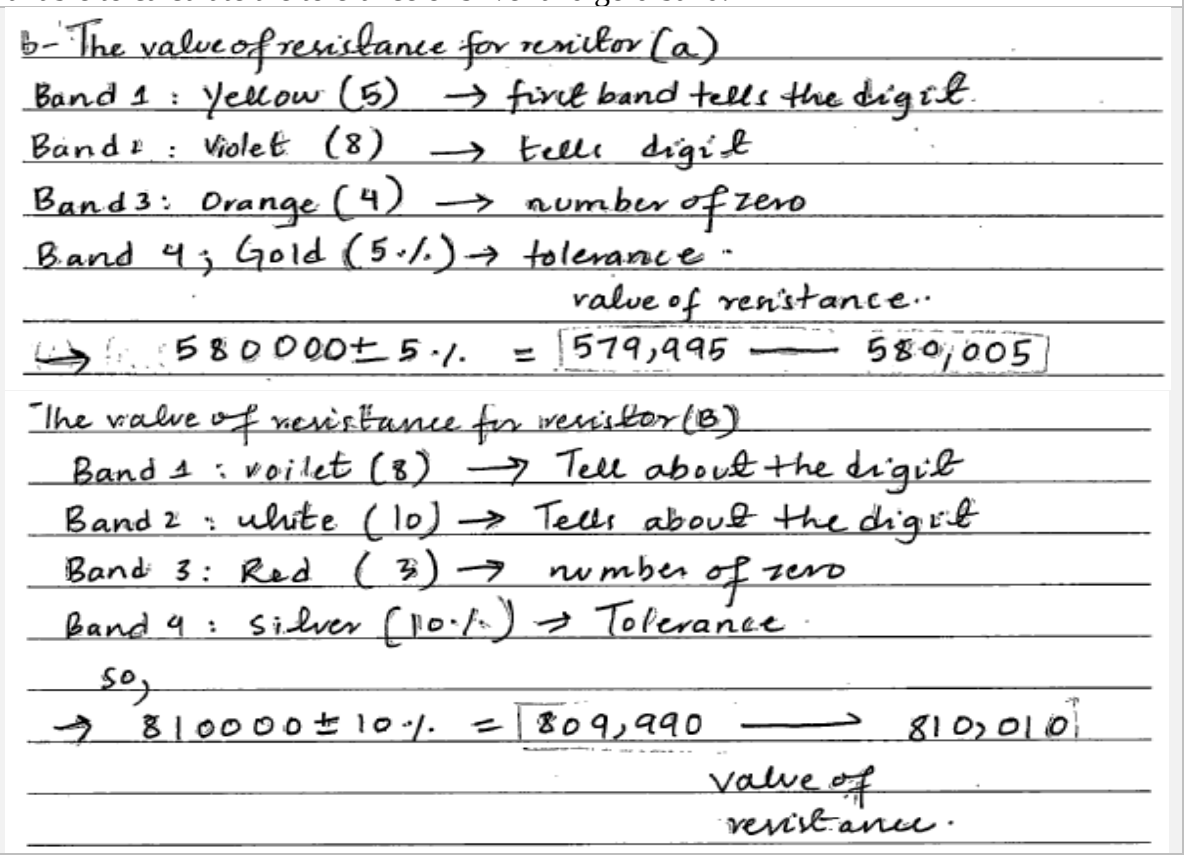
Suggestions for improvement (Highlight all that apply)

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Any Additional Suggestion:

- Providing more practice with Kirchhoff's laws and circuit analysis, especially in complex circuits involving internal resistance, can strengthen candidates' understanding and skills in this area.
- Reinforcing the proper application of Kirchhoff's law will further enhance accuracy and confidence in solving circuit problems.
- Clarifying the distinction between e.m.f. and terminal voltage, particularly in the context of circuits with internal resistance is crucial.

<table>
<tr><th colspan="2">Question No. 9b</th></tr>
<tr><td>Question Text</td><td>Two colour code resistors exhibit the following colours.
<table><tr><th>Resistor (a)</th><th>Resistor (b)</th></tr><tr><td>Band 1: Yellow</td><td>Band 1: Violet</td></tr><tr><td>Band 2: Violet</td><td>Band 2: White</td></tr><tr><td>Band 3: Orange</td><td>Band 3: Red</td></tr><tr><td>Band 4: Gold</td><td>Band 4: Silver</td></tr></table>
Find the value of resistance for each resistor.</td></tr>
<tr><td>SLO No.</td><td>12.3.6</td></tr>
<tr><td>SLO Text</td><td>Calculate the value of carbon resistance by using colour codes.</td></tr>
<tr><td>Max Marks</td><td>07</td></tr>
<tr><td>Cognitive Level</td><td>A</td></tr>
<tr><td>Checking Hints</td><td>a.
1 mark for the correct coding.
1 mark for the correct substitution.
1 mark for finding the resistance.
b.
1 mark for the correct coding.
1 mark for the correct substitution.
1 mark for finding the resistance.
1 mark for writing the correct tolerance in both parts</td></tr>
<tr><td>Overall Performance</td><td>The majority of candidates attempted this part of the question and showed a strong understanding of how to decode carbon resistor values using the colour code. Even without the colour code table provided, they created their table and accurately calculated the resistance value along with its tolerance range.</td></tr>
<tr><td>Description of Better Responses</td><td>Better responses demonstrated a clear understanding of the significance of the first, second, third and fourth colour bands in carbon resistors. Candidates effectively decoded the resistance values from the colour bands as 47000 and 79000 and accurately determined the tolerance levels as 5% and 10% respectively. Many responses included detailed workings, with candidates creating their colour code tables and calculating not only the resistance values but also the corresponding tolerance ranges.</td></tr>
</table>

Image of Better Response	
Description of Weaker Responses	In *weaker responses*, candidates faced difficulties in accurately decoding the resistor values due to limited familiarity with the meanings of the first, second, third and fourth colour bands, which led to miscalculations in both parts. They wrongly named the colour code for each colour of the resistor. Their mathematical manipulation and articulation in finding the correct value of carbon resistance by using colour codes were very weak. They were also unable to calculate the tolerance of silver and gold band.
Image of Weaker Response	

Suggestions for improvement (Highlight all that apply)

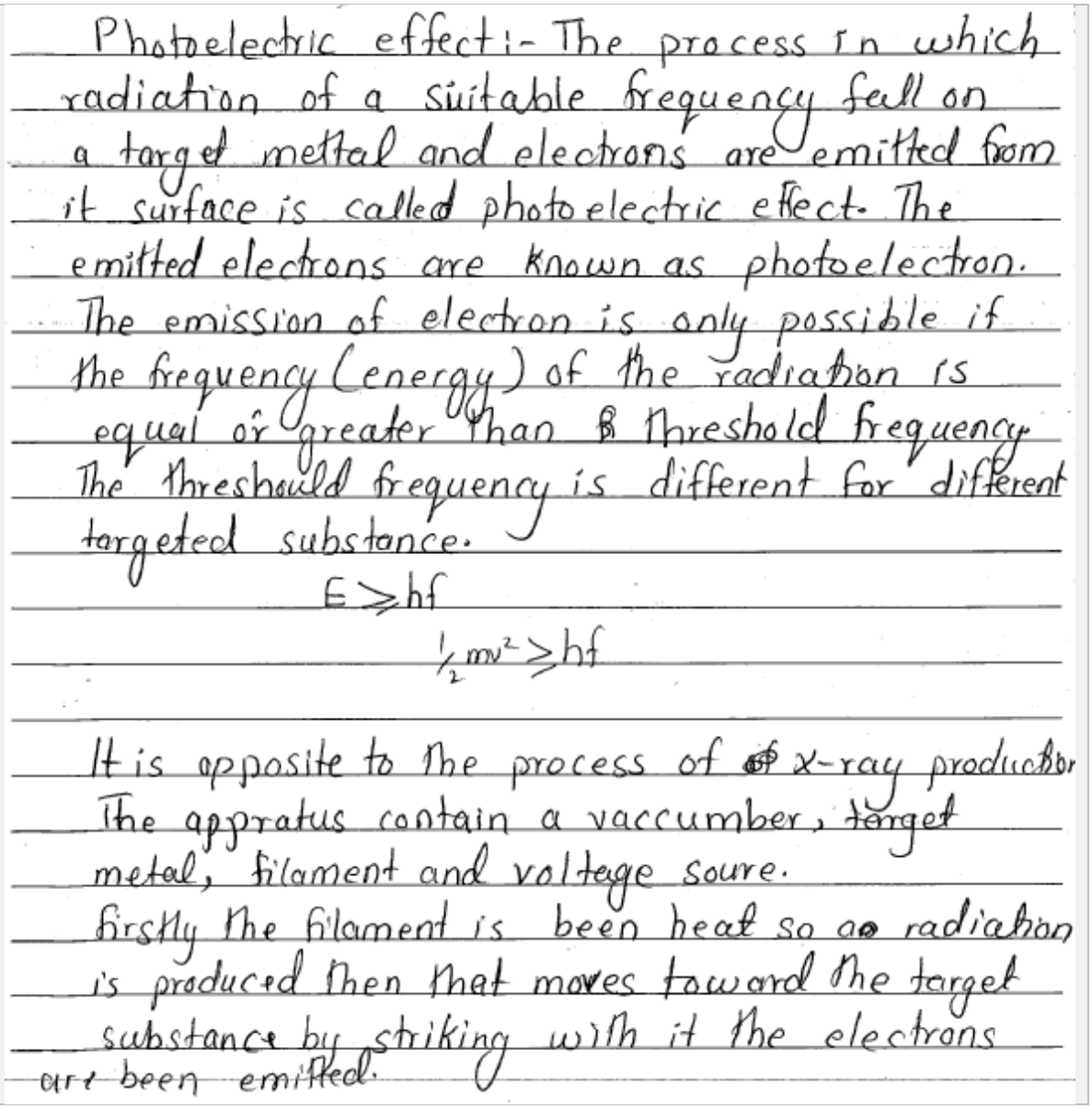
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Any Additional Suggestion:

- Guiding these candidates to acquaint themselves with the colour code and its corresponding values can address this concern.
- Teachers should emphasise the significance of resistor band interpretation and encouraging correct decoding will likely enhance their accuracy.
- Constructive feedback by the subject teachers, coupled with focused practice, will augment their foundational understanding and promote more precise calculations in analogous scenarios.

Question No. 10a	
Question Text	Describe Albert Einstein's explanation about photoelectric effect and its photoelectric equation.
SLO No.	18.3.3
SLO Text	Derive Einstein's photoelectric equation.
Max Marks	07
Cognitive Level	U
Checking Hints	1 mark for each mentioned point (4 required) 1 mark for each highlighted statement in the photoelectric equation (3 required)
Overall Performance	The majority of the candidates attempted this part of the question and accurately fulfilled the demand of the question, evidencing a strong grasp of the photoelectric effect and Einstein's explanation. Many candidates accurately highlighted that light consisted of photons and that the energy of these photons is quantised, being proportional to their frequency and explained Einstein's explanation by providing the correct equation. Overall, while many candidates demonstrated a good understanding of Einstein's photoelectric effect theory and associated equations, there is a need for clearer explanations of the concepts and classical failures.

Description of Better Responses	*Better responses* demonstrated a strong understanding of the photoelectric effect and Einstein's explanation and classical failures. Most of the candidates correctly explained the photoelectric effect and photoelectric equation $E = hf_0 - \Phi$. Candidates' responses effectively discussed the classical failures, such as the inability of the wave theory of light to explain the immediate emission of electrons regardless of the intensity of light and the observation that light below a certain frequency does not emit electrons, regardless of its intensity. The distinction between intensity and frequency in the context of the photoelectric effect is crucial. Emphasising that the kinetic energy of the emitted electrons depends on the frequency of the incident light, not its intensity will clarify these concepts.
Image of Better Response	• Albert Einstein put forward his explanation regarding photoelectric effect as when a photon collides with a metal surface it cause electrons to get released from the metal surface. • These electrons are called photoelectrons. Due to their negative nature they get attracted towards anode (+ve) constituting photoelectric current. • The energy of a photon is converted into the work function and kinetic energy of the electron. $K.E = hf - hf_0$ $hf = \phi + K.E\ (\bar{e})$ or $hf = hf_0 + K.E\ (\bar{e})$ • The minimum frequency to release an electron from the metal surface is known as Threshold frequency (f_0) • In order to find out the maximum energy of an electron the polarity of anode and cathode are reversed ie anode is given negative potential while cathode is positive. • Due to anode being negative is repels electron as electrons are also negative in nature. This when increased significantly causes photoelectric current to become 0. The voltage point at which the current becomes zero is known as stopping potential (V_0) $\frac{1}{2}mv^2 = eV_0$ • Photoelectric effect depicts the particle nature of photon. • The No. of electrons depends upon the intensity of photon. • The kinetic energy of electrons depend upon the frequency of photon
Description of Weaker Responses	*In weaker responses*, some candidates exhibited misconceptions or misinterpreted the question, leading to inaccurate explanations. Incorrect or incomplete explanations were also provided. A few candidates confused key concepts, such as incorrectly attributing the energy dependence to the intensity of light rather than its frequency. This confusion resulted in incorrect explanations of the photoelectric effect and the associated equation as mentioned in the better responses. Additionally, some responses lacked clarity or completeness in addressing the failures of classical theory, either by omitting them entirely or by failing to adequately explain why the classical wave theory could not account for the observed phenomena.

Image of Weaker Response	

Photoelectric effect:- The process in which radiation of a suitable frequency fall on a target mettal and electrons are emitted from it surface is called photo electric effect. The emitted electrons are known as photoelectron. The emission of electron is only possible if the frequency (energy) of the radiation is equal or greater than a threshold frequency. The threshould frequency is different for different targeted substance.

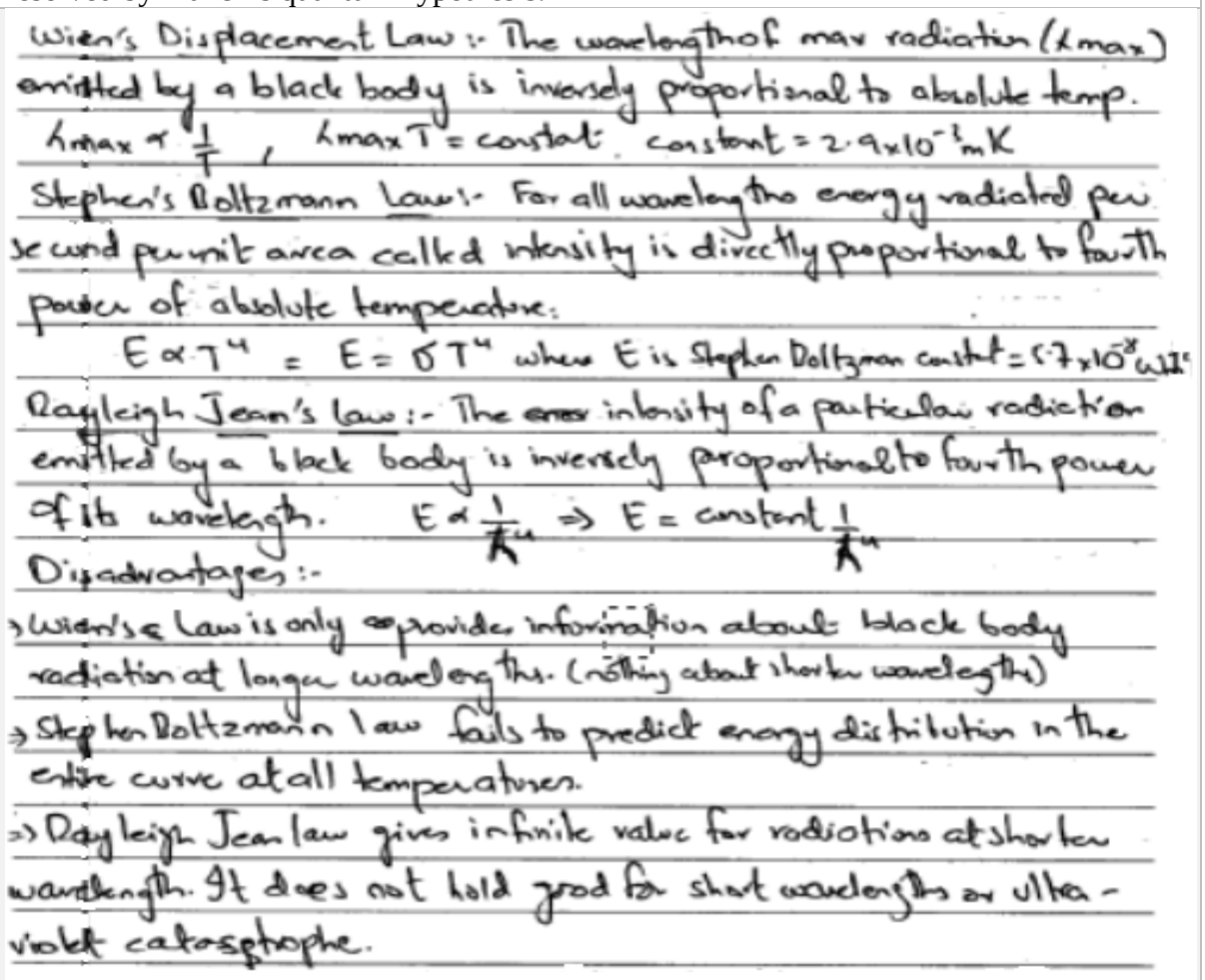
$E \geq hf$

$\frac{1}{2}mv^2 \geq hf$

It is opposite to the process of x-ray production. The appratus contain a vaccumber, ~~target~~ metal, filament and voltege soure.
firstly the filament is been heat so as radiation is produced then that moves toward the target substance by striking with it the electrons are been emitted.

Suggestions for improvement (Highlight all that apply)

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Any Additional Suggestion:		

Question No. 10b	
Question Text	Describe the laws governing black body radiation and explain its FOUR disadvantages.
SLO No.	18.2.2
SLO Text	Describe laws governing blackbody radiations and their drawbacks.
Max Marks	07
Cognitive Level	U
Checking Hints	1 mark for writing each highlighted statement in Planck's Law (2 required) 1 mark for writing Stefan-Boltzmann Law 1 mark for each mentioned disadvantage (4 required)
Overall Performance	The overall performance on this question was positive. Candidates accurately addressed the question, clearly understanding the black body radiation. Many demonstrated a good grasp of the fundamental laws governing black body radiation and their limitations. However, there is room for improvement in explaining the details and limitations of these laws.
Description of Better Responses	*Better responses* effectively explained the key laws governing black body radiation, including Wien's displacement law, Rayleigh-Jean's law, Stefan-Boltzmann law and Plank's law. These responses demonstrated a clear understanding of these principles, particularly how Plank's law describes the spectral distribution of radiation. Additionally, they accurately addressed the limitations of these laws, such as the failure of classical theories like Rayleigh-Jeans law to predict the observed spectrum, leading to the ultraviolet catastrophe, which was resolved by Planck's quantum hypothesis.
Images of Better Responses	
Description of Weaker Responses	In *weaker responses*, few candidates confused the laws or provided incomplete explanations, particularly in differentiating between various laws and their specific implications. This often led to unclear or incorrect descriptions of the phenomena they govern. Additionally, a few candidates did not adequately address the limitations of these laws, either by omitting them

	entirely or by failing to clearly explain why classical theories could not accurately describe black body radiation. They wrote about the absorption and emission of radiation and while writing the drawbacks, they were unable to mention ultraviolet catastrophe, quantisation, complexity and idealisation factors.
Image of Weaker Response	b. Black body ~~radio~~ is an ideal body that can absorb all the radiations falling on it. Black body consist of blackened body that make it good absorber. It consist of a hole from which radiations can enter and leave. 1st law governing black body radiations is that the radiations have wave like nature that interacts with matter. 2/ Radiations are absorbed by the substance depending upon the color exhibiting by that substance. 3/ The extent of the absorption of radiation by substance depends upon the substance absorbing quality. * Disadvantages. 1/ By absorbing all the radiation the black body can increase the temperature 2/ Person adjusting the black body can experience the absor exposure to the radiations and face harmful affects. 3/ Black body after absorbing the radiations emits it into the space cause harm to environment.

Suggestions for improvement (Highlight all that apply)

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Any Additional Suggestion:

- Teachers may classify the distinctions and specific contributions of each law governing black body radiation, along with their respective formulas and implications will help candidates better understand and explain these concepts.
- Subject teachers must emphasise the historical context and the specific problems that Plank's theory addressed, particularly the ultraviolet catastrophe, can provide a deeper understanding of why classical physics failed and how quantum theory provided solutions.

Annexure A: Pedagogies Used for Teaching the SLOs

Pedagogy: Storyboard
Description: A visual pedagogy that uses a series of illustrated panels to present a narrative, encouraging creativity and critical thinking. It helps learners organise ideas, sequence events, and comprehend complex concepts through storytelling.
Example: In a Literature class, students are tasked with creating storyboards to visually retell a novel. They draw key scenes, write captions, and present their stories to the class, enhancing their reading comprehension and fostering their imagination.

Pedagogy: Cause and Effect
Description: This pedagogy explores the relationships between actions and consequences. By analysing cause-and-effect relationships, learners develop a deeper understanding of how events are interconnected and how one action can lead to various outcomes.
Example: In a History class, students study the causes and effects of the Industrial Revolution. They research and discuss how technological advancements in manufacturing led to significant societal changes, such as urbanisation and labour reform movements.

Pedagogy: Fish and Bone
Description: A method that breaks down complex topics into main ideas (the fish) and supporting details (the bones). This visual approach enhances comprehension by highlighting essential concepts and their relevant explanations.
Example: During a Biology class on human anatomy, the teacher uses the fish and bone technique to teach about the human skeletal system. Teacher presents the main components of the human skeleton (fish) and elaborates on each bone's structure and function (bones).

Pedagogy: Concept Mapping
Description: An effective way to visually represent relationships between ideas. Learners create diagrams connecting key concepts, aiding in understanding the overall structure of a subject and fostering retention.
Example: In a Psychology assignment, students use concept mapping to explore the various theories of personality. They interlink different theories, such as Freud's psychoanalysis, Jung's analytical psychology, and Bandura's social-cognitive theory, to see how they relate to each other.

Pedagogy: Audio Visual Resources
Description: Incorporating multimedia elements like videos, images, and audio into lessons. This approach caters to different learning styles, making educational content more engaging and memorable.
Example: In a General Science class, the teacher uses a documentary-style video to teach about the solar system. The video includes stunning visual animations of the planets, interviews with astronomers, and background music, enhancing students' interest and understanding of space.

Pedagogy: Think, Pair, and Share
Description: A collaborative learning technique where students ponder a question or problem individually, then discuss their thoughts in pairs or small groups before sharing with the entire class. It fosters active participation, communication skills, and diverse perspectives.
Example: In a Literature in English class, the teacher poses a thought-provoking question about a novel's moral dilemma. Students first reflect individually, then pair up to exchange their opinions, and finally participate in a lively class discussion to explore different viewpoints.

Pedagogy: Questioning Technique (Socratic Approach)
Description: Based on Socratic dialogue, this method stimulates critical thinking by posing thought-provoking questions. It encourages learners to explore ideas, justify their reasoning, and discover knowledge through a process of inquiry.
Example: In an Ethics class, the instructor uses the Socratic approach to lead a discussion on the meaning of justice. By asking a series of probing questions, the students engage in a deeper exploration of ethical principles and societal values.

Pedagogy: Practical Demonstration
Description: A hands-on approach where learners observe real-life applications of theories or skills. Practical demonstrations enhance comprehension, skill acquisition, and problem-solving abilities by bridging theoretical concepts with real-world scenarios.
Example: In a Food and Nutrition class, the instructor demonstrates the proper technique for filleting a fish. Students observe and then practice the skill themselves, learning the practical application of knife skills and culinary precision.

(**Note:** The examples provided in this annexure serve as illustrations of various pedagogies. It is important to understand that these pedagogies are versatile and can be applied across subjects in numerous ways. Feel free to adapt and explore these techniques creatively to enhance learning outcomes in your specific context.)

Acknowledgements

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Additionally, we express our gratitude to the esteemed team of reviewers for their constructive feedback on overall performance, better and weaker responses, and validating teaching pedagogies along with suggestions for improvement.

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